

統計計算

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Monty Hall Problem



圖片來源: <https://wirelesspi.com/the-reason-why-the-monty-hall-problem-continues-to-perplex-everyone/>

Outline

1

Introduction and Problem Setup

Monty Hall Problem、Simulation Framework

2

Statistical Inference

- Bootstrap
- Jackknife Method
- Permutation Test
- Markov Chain Monte Carlo (MCMC)-Metropolis Hastings Algorithm
- Acceptance-Rejection

3

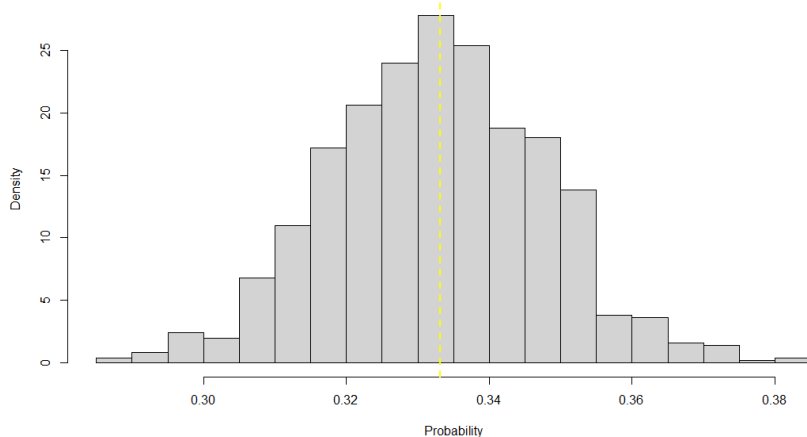
Explanation

Original Version of the Monty Hall Problem

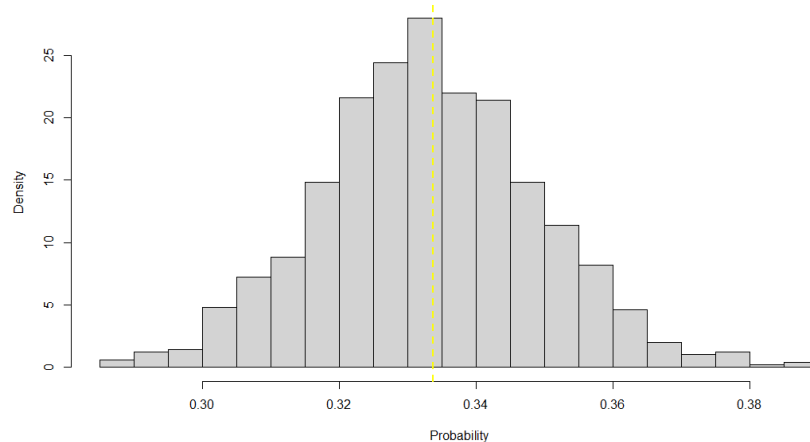
When contestant selected the sport car at the 1st time

- 1000個人
- 每個人玩了1000次

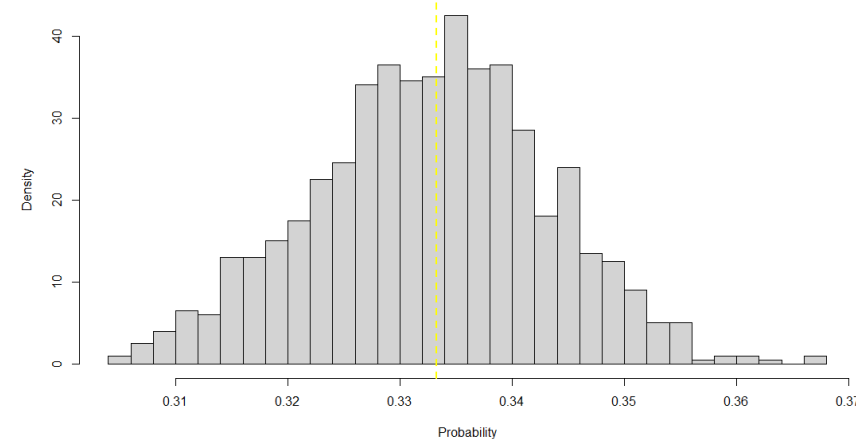
Distribution of Sampling (R Bernoulli Simulation)



Distribution of Sampling (Inverse Function)



Distribution of Sampling (Inverse Function with Control Variate)



Inverse function

$$X \sim \text{Bernoulli}\left(\frac{1}{3}\right)$$

$$U \sim \text{Uniform}(0, 1)$$

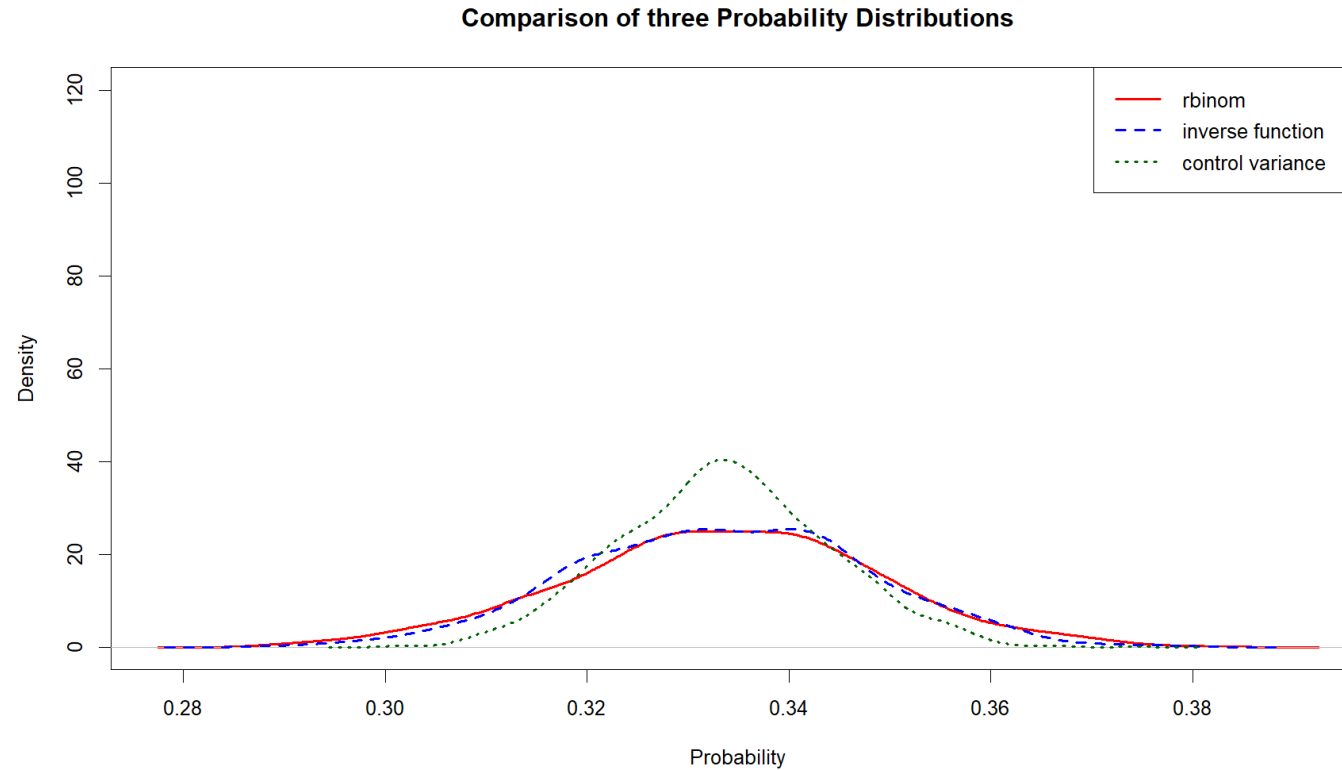
$$X = \begin{cases} 1, & U \leq \frac{1}{3} \\ 0, & U > \frac{1}{3} \end{cases}$$

antithetic variates

$$U \sim \text{Uniform}(0, 1) \quad X = \begin{cases} 1, & U \leq \frac{1}{3} \\ 0, & U > \frac{1}{3} \end{cases}$$
$$U' = 1 - U \quad X' = \begin{cases} 1, & U' \leq \frac{1}{3} \\ 0, & U' > \frac{1}{3} \end{cases}$$

$$\hat{\theta}_{AV} = \frac{X + X'}{2}$$

When contestant selected the sport car at the 1st time



Variance reduction = 47.68226 %

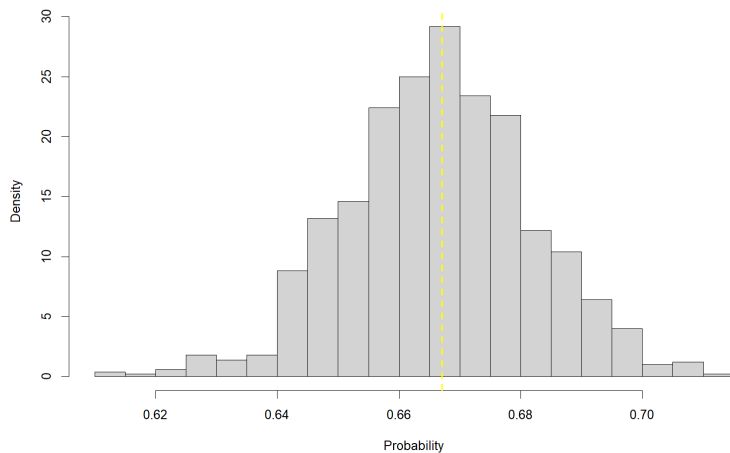
Method	Mean	Var
Built in R Bernoulli Simulation	0.333122	0.000230922
Inverse Function	0.333221	0.0002211353
Inverse Function with Control Variance	0.333348	0.0001184333

/ Second choice

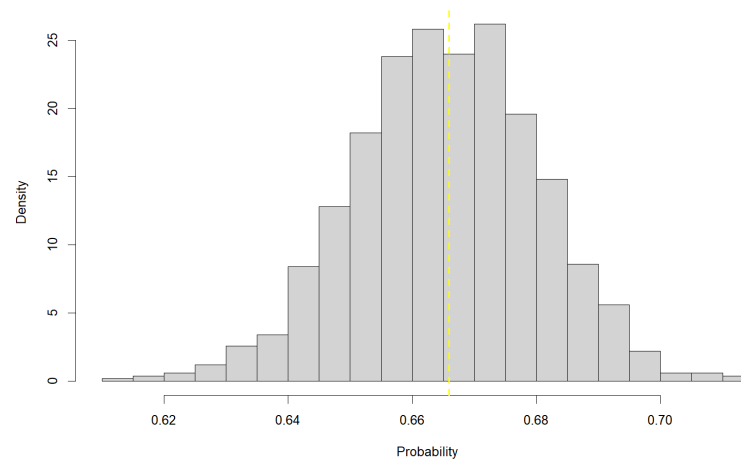
a.If host **knows** the correct answer:

$$\text{換門後是否贏得跑車} = \begin{cases} 1, & \text{第一次沒選到跑車} \\ 0, & \text{第一次選到跑車} \end{cases}$$

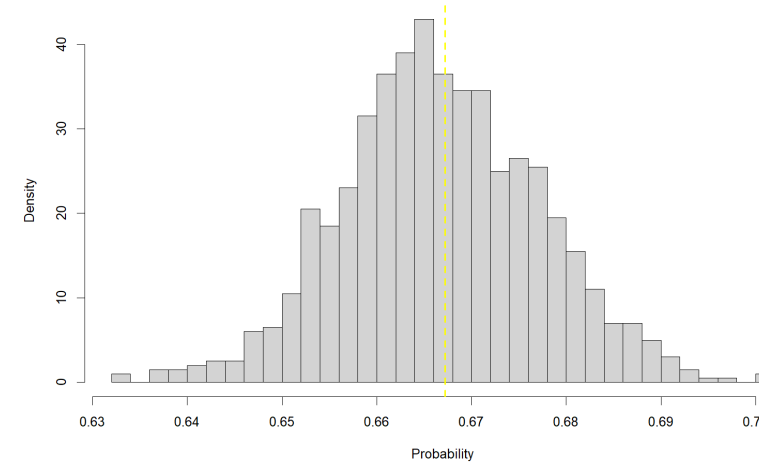
Distribution of Sampling (R Bernoulli Simulation)



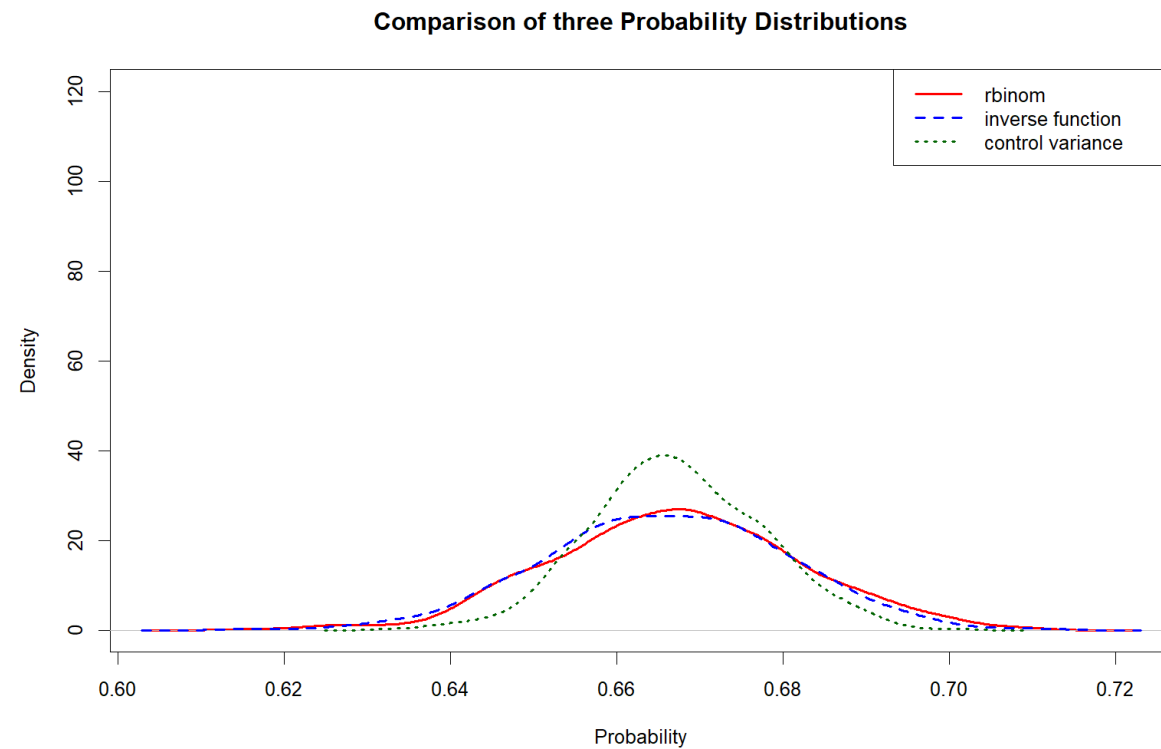
Distribution of Sampling (Inverse Function)



Distribution of Sampling (Inverse Function with Control Variance)



/ Second choice



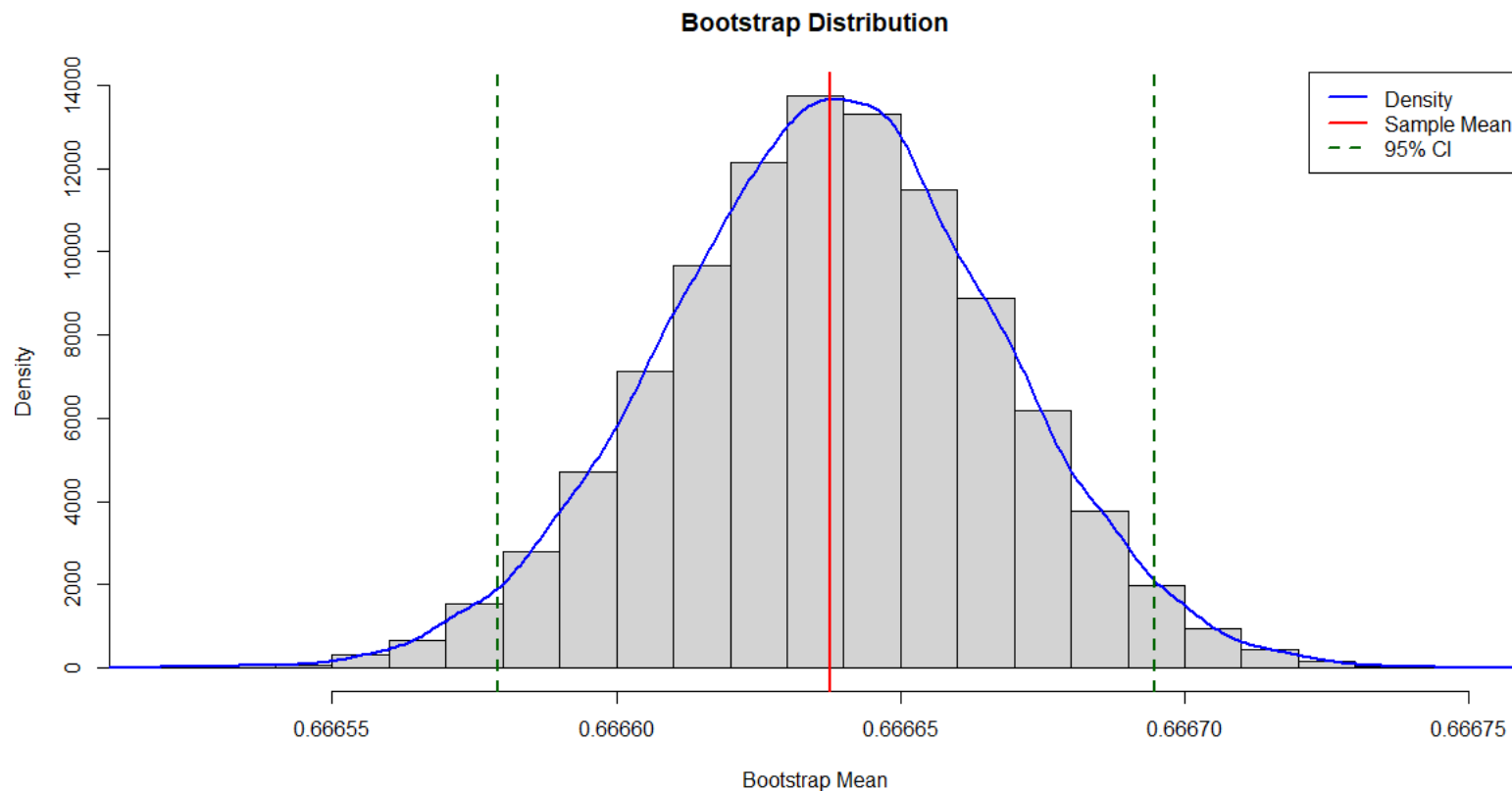
Variance reduction = 46.61676%

Method	Mean	Var
Built in R Bernoulli Simulation	0.667405	0.0002047159
Inverse Function	0.666227	0.0002161997
Inverse Function with Control Variance	0.667471	0.0001154144

100個人
每個人玩了1000次
每次都換門的勝率



抽10000次

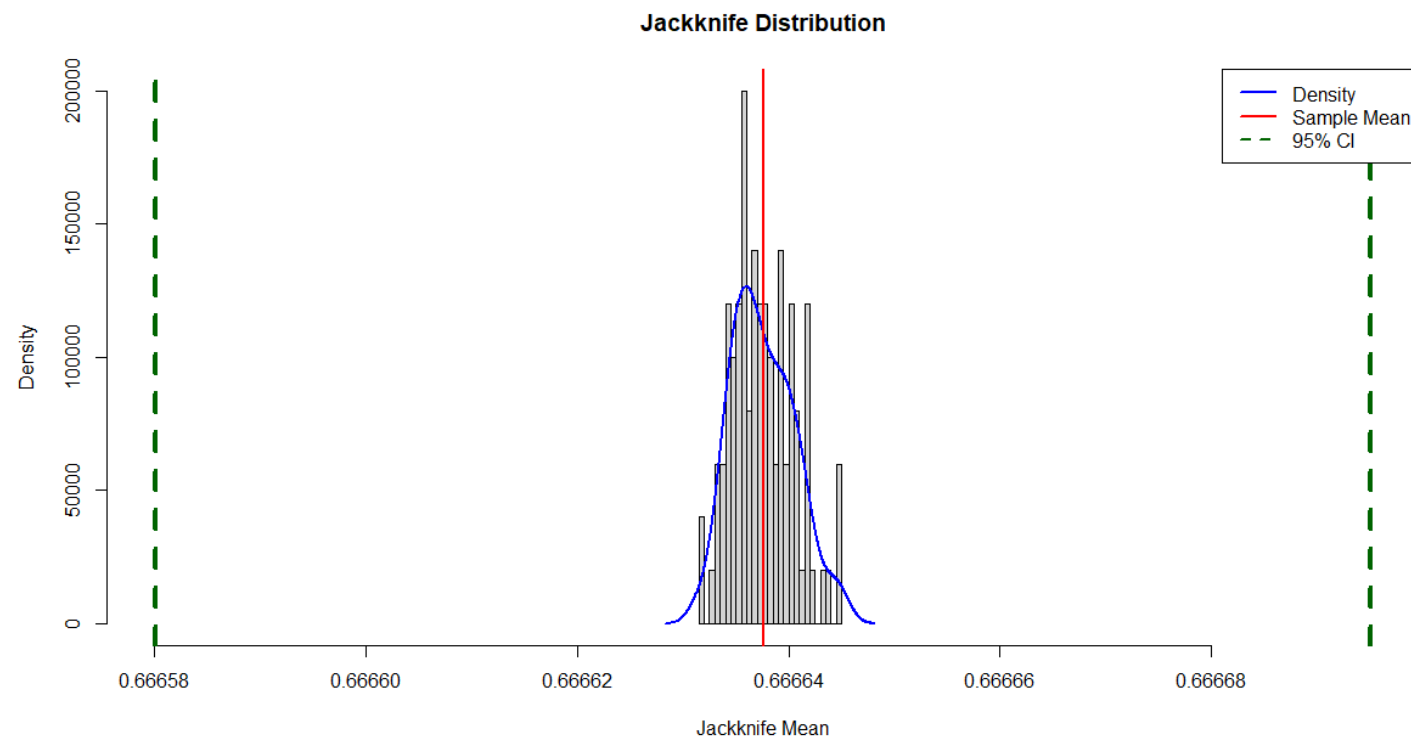


Method	mean	bias	std. error
Bootstrap	0.6666376	0.000000126376	0.0000294129

100個人
每個人玩了1000次
每次都換門的勝率

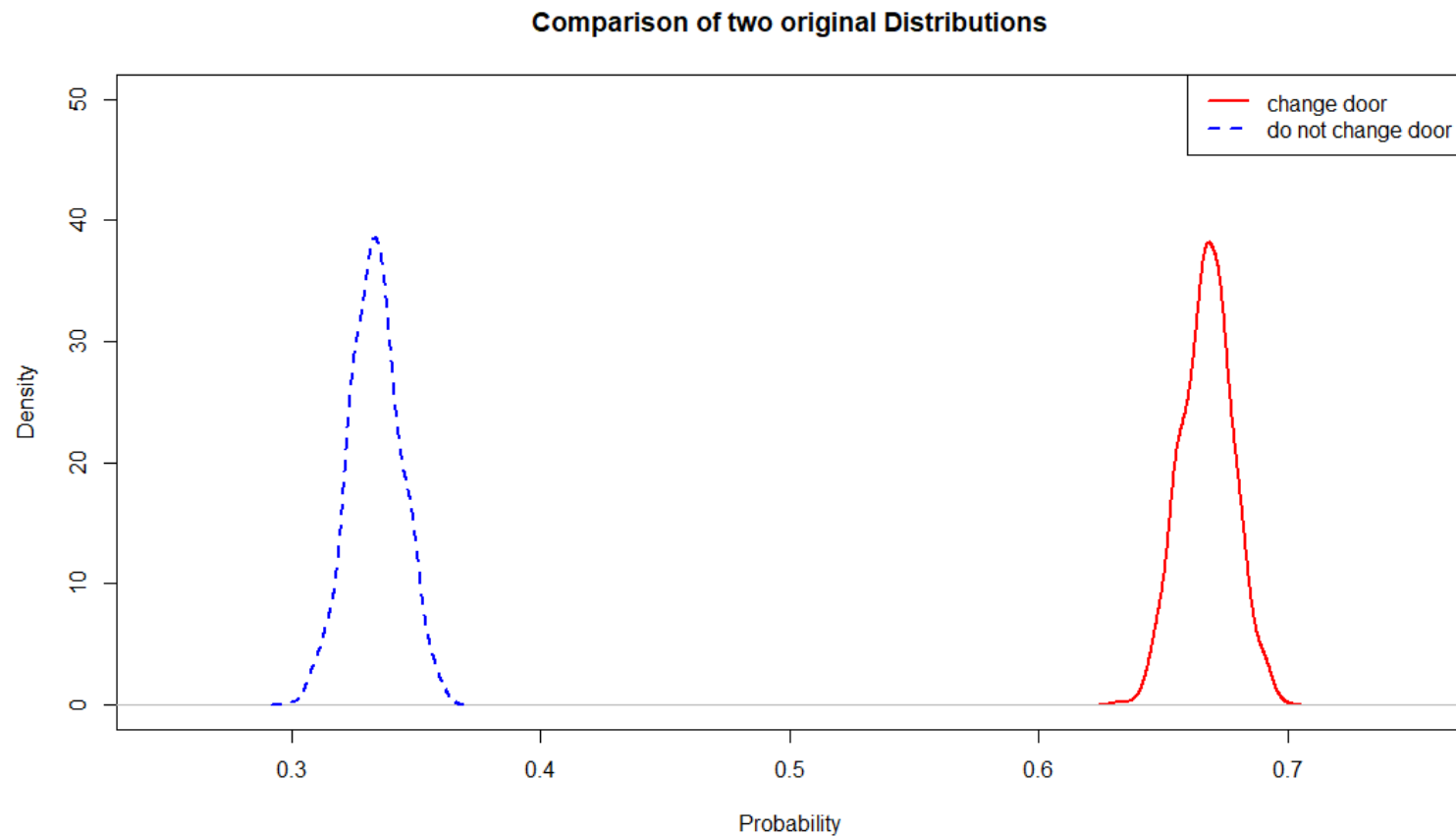


每次drop掉1人



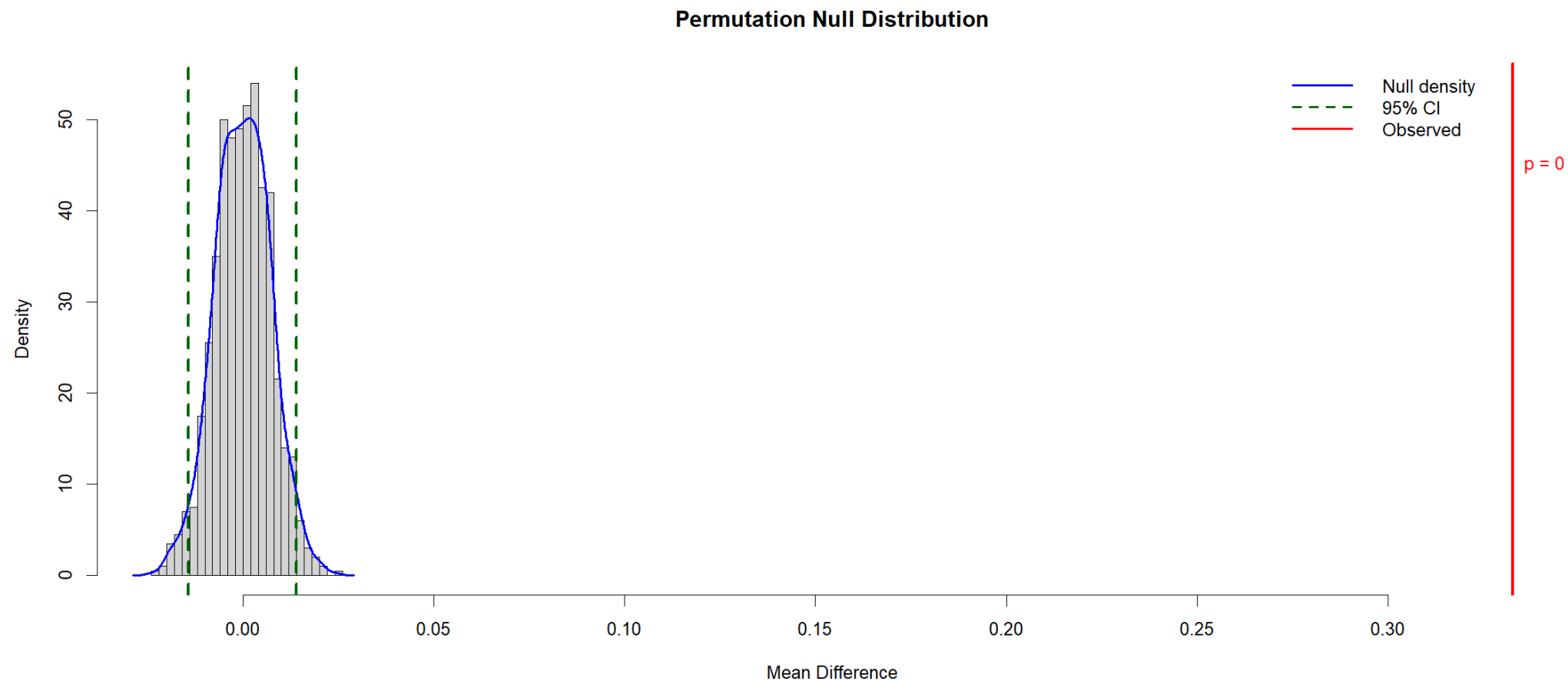
Method	mean	std. error
Jackknife	0.6666376	0.00002934132

1000個人
每個人玩了1000次



不換門後結果 = $\begin{cases} 0, & \text{第一次選到羊} \\ 1, & \text{第一次選到車} \end{cases}$

換門後是否贏得跑車 = $\begin{cases} 1, & \text{第一次沒選到跑車} \\ 0, & \text{第一次選到跑車} \end{cases}$



做了1000次

Method	mean	std. error	observed mean
Permutation	-0.00006004	0.007290593	0.332783

1個人玩了1000次
每次都換門的結果

$$k = \sum_{i=1}^n y_i$$

$$P(\theta | y) \propto P(y | \theta)P(\theta)$$

一般人相信換不換門勝率一樣

$$\theta^k (1 - \theta)^{n-k}$$

$$\theta^{a-1} (1 - \theta)^{b-1}$$

Beta(10, 10)

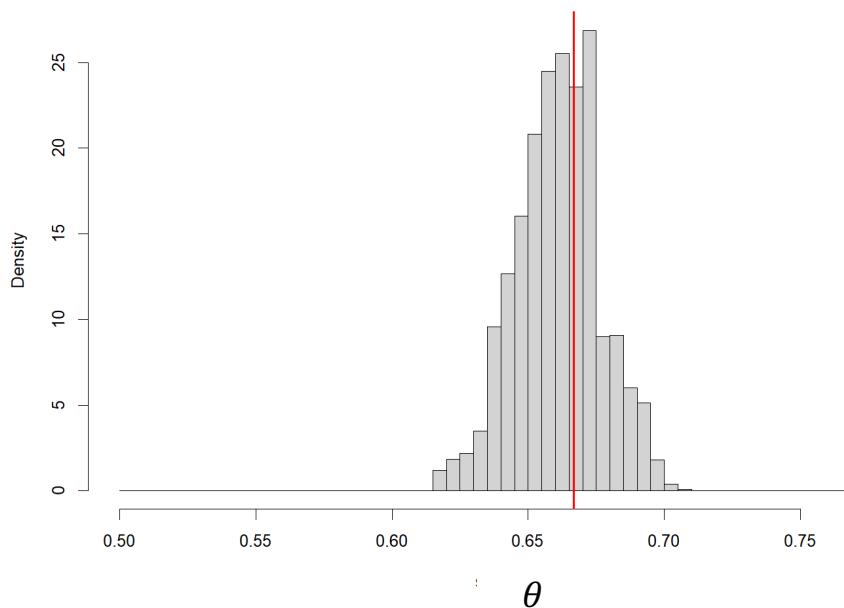
$$E[\theta] = \frac{10}{10 + 10} = \frac{1}{2}$$

$$\theta^{k+a-1} (1 - \theta)^{n-k+b-1}$$

一開始認為換不換門勝率一樣，設定0.5

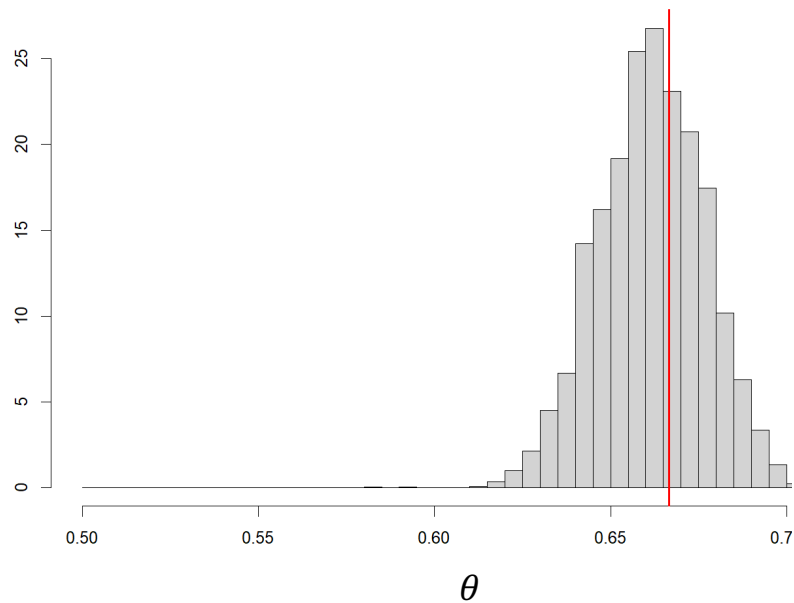
$$\theta_{\text{prop}} = \theta_{\text{current}} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

Histogram of samples



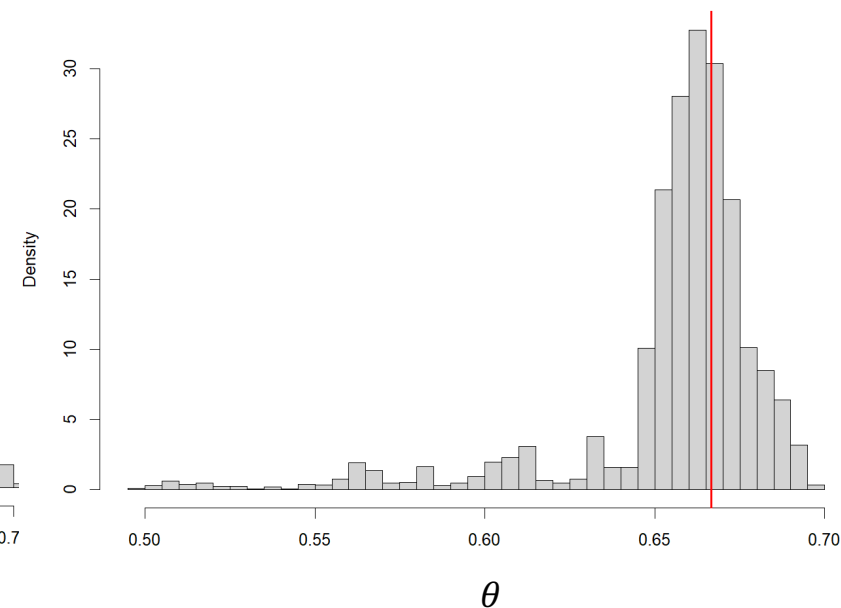
$\sigma = 0.5$

Histogram of samples



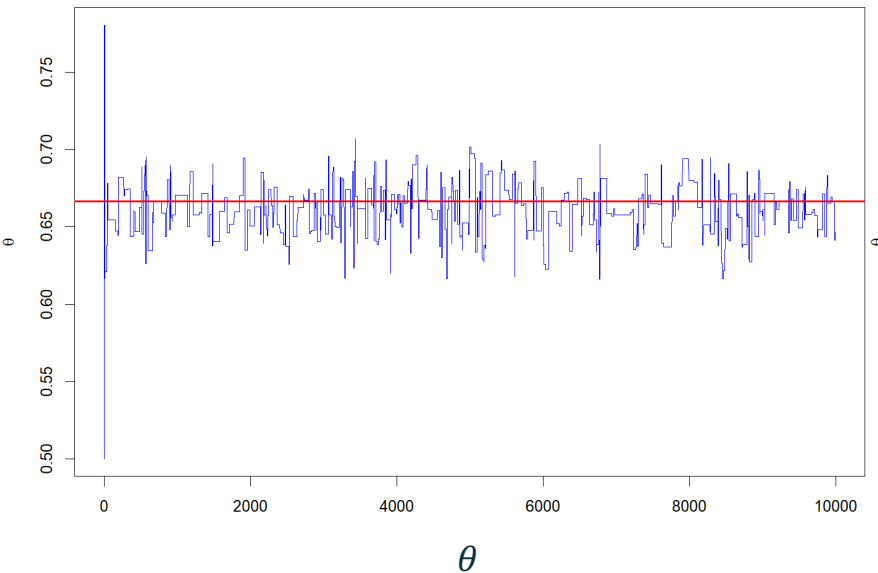
$\sigma = 0.05$

Histogram of samples

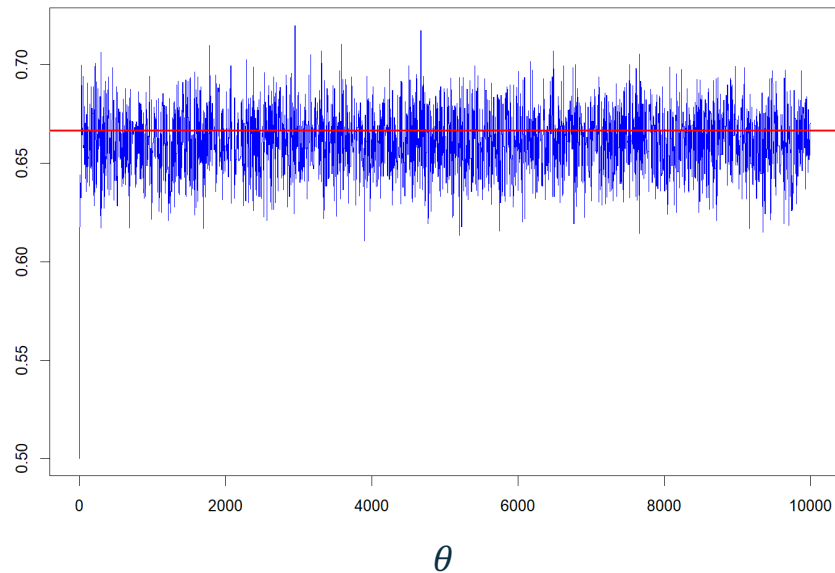


$\sigma = 0.001$

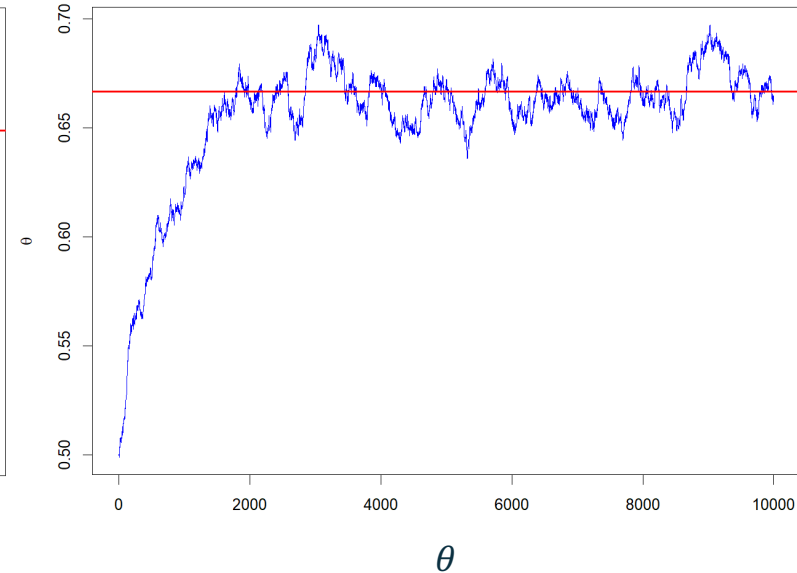
MH Trace Plot

 $\sigma = 0.5$

MH Trace Plot

 $\sigma = 0.05$

MH Trace Plot

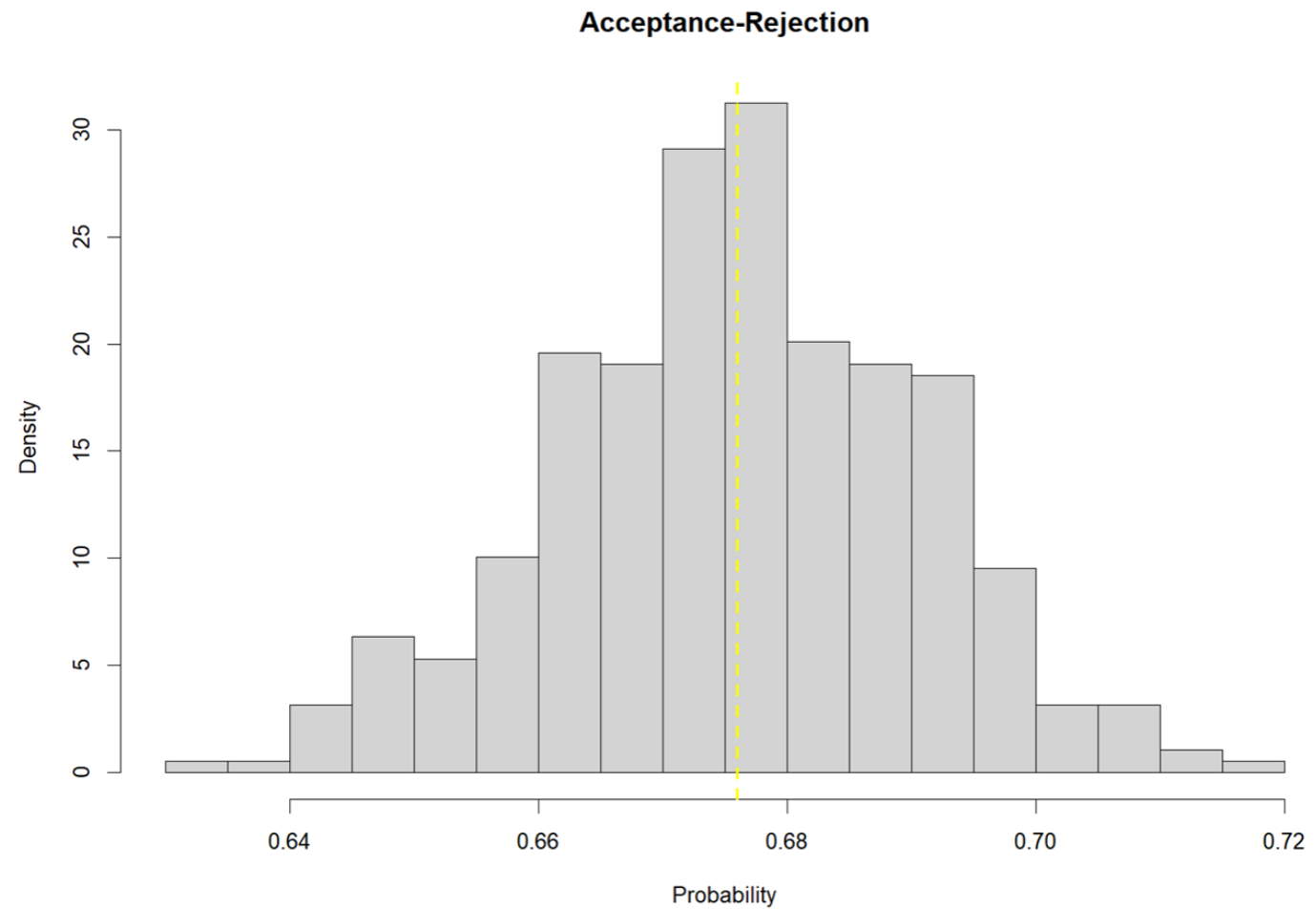
 $\sigma = 0.001$

2 Statistical Inference: Acceptance - Rejection

$$g(t) \sim \text{Uniform}(0, 1)$$

$$\frac{f(t)}{g(t)} \leq c$$

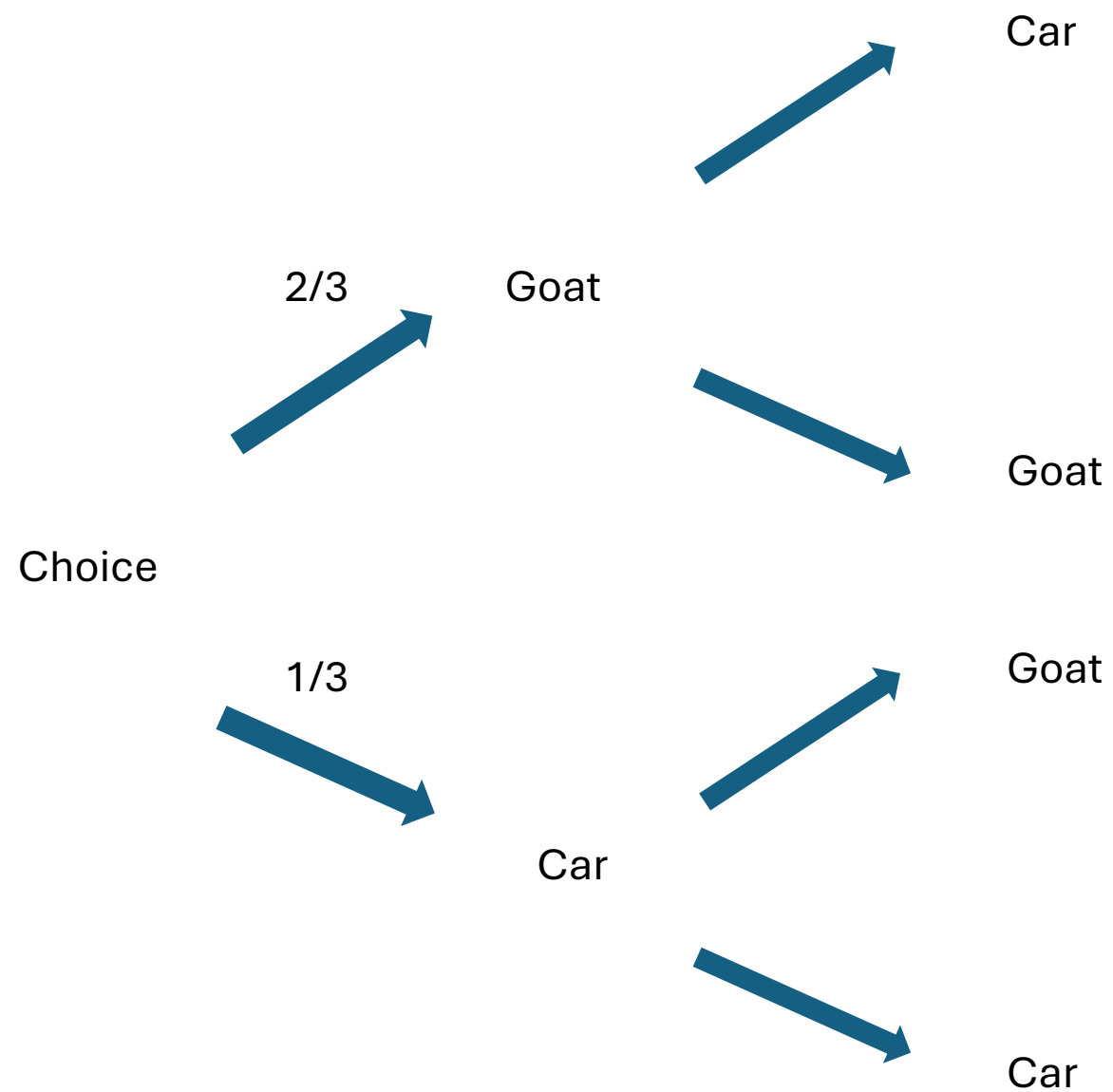
$$N=10000$$





Explanation

$$P_{car} = 2/3$$



What if host lost the information (The situation as commonly imagined)

/

Why we can use inverse function + antithetic variates in second choice

Inverse function + antithetic variates

$$Y_i = \begin{cases} 0, & X_i = 1 \\ Z_i, & X_i = 0, \end{cases} \quad Z_i \sim \text{Bernoulli} \left(\frac{1}{2} \right)$$

主持人開出跑車，遊戲沒辦法玩

主持人開出羊，換門後贏得跑車

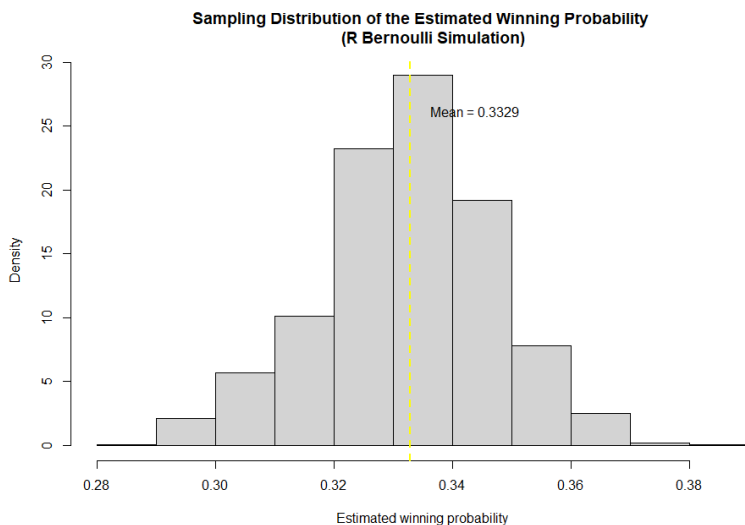
$$X_i = \begin{cases} 1, & \text{第一次選到跑車} \\ 0, & \text{第一次沒選到跑車} \end{cases}$$

/ First and second choice

1000個人
每個人玩了1000次
每次都換門的勝率

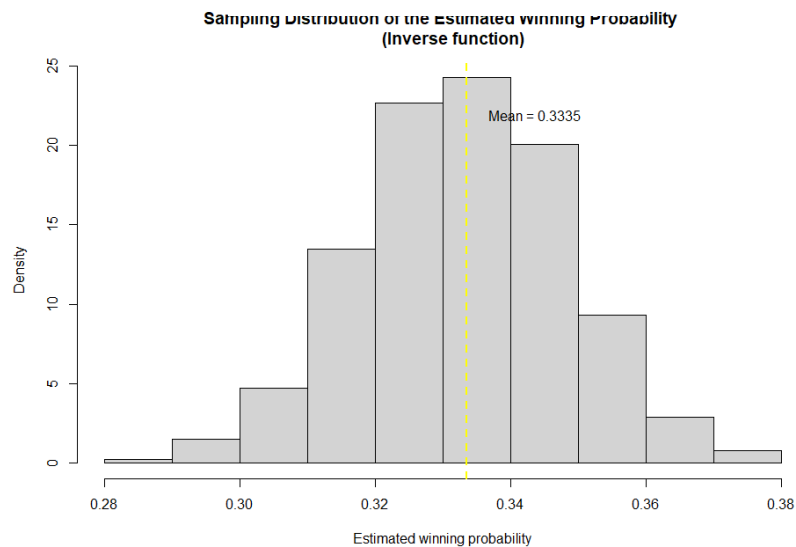
$$X \sim \text{Bernoulli}\left(\frac{1}{3}\right)$$

$$X \sim \text{Bernoulli}\left(\frac{1}{2}\right)$$



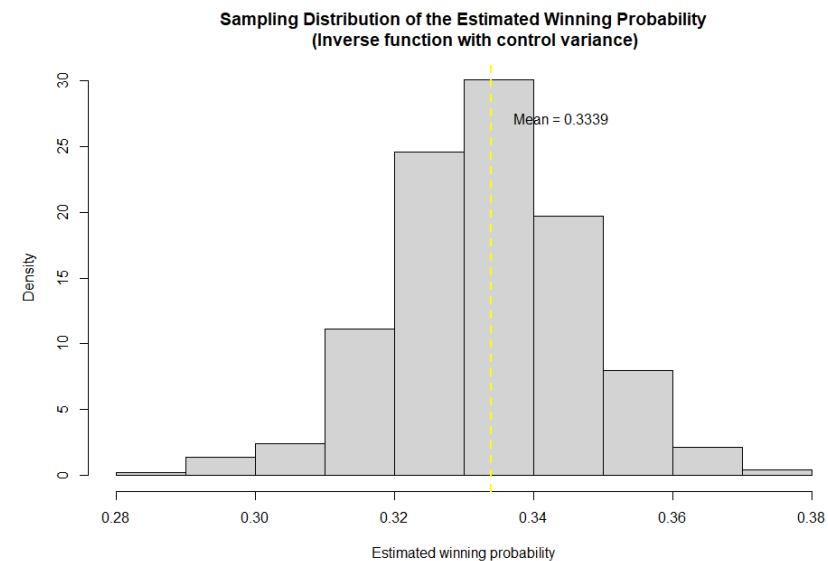
Inverse function

$$X \sim \text{Bernoulli}\left(\frac{1}{2}\right)$$

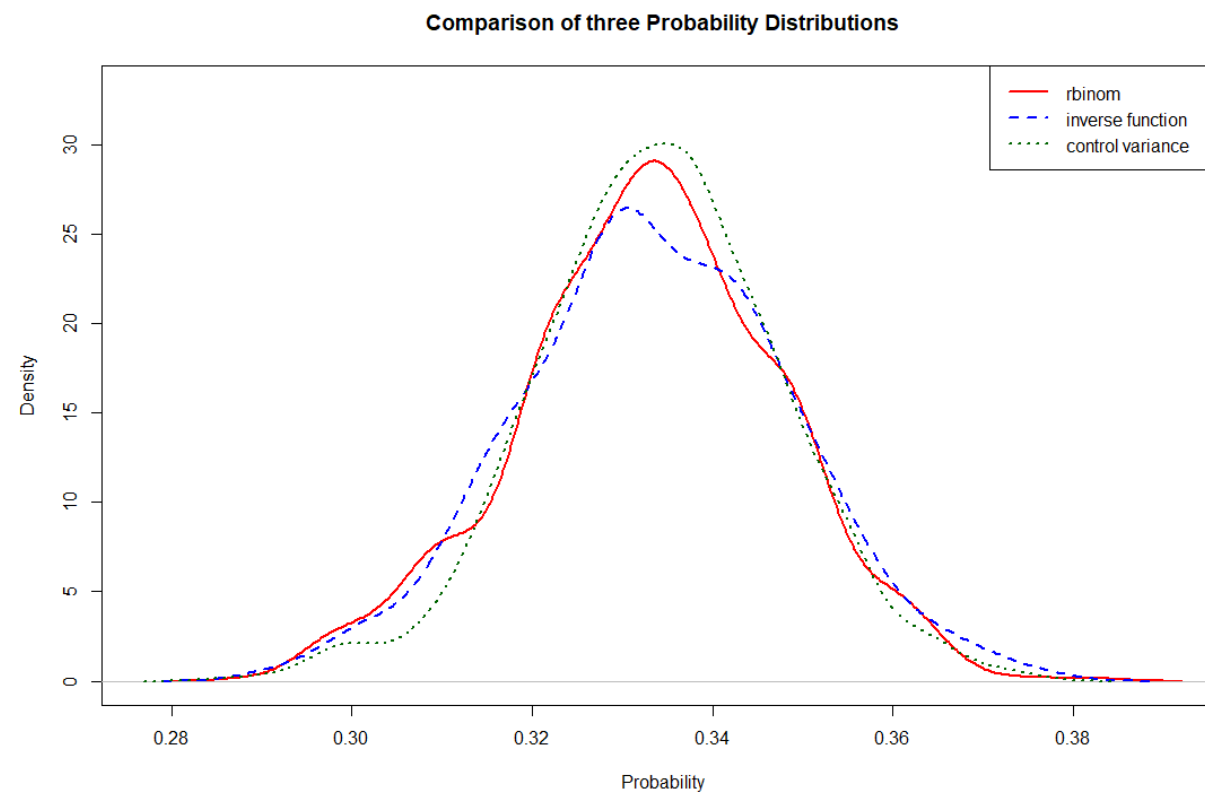


Antithetic variates

$$X \sim \text{Bernoulli}\left(\frac{1}{2}\right)$$



Comparison



Variance reduction = 21.52862%

Method	Mean	Var
Built in R Bernoulli Simulation	0.332871	0.0002073998
Inverse Function	0.333238	0.0002334708
Inverse Function with Control Variance	0.333604	0.0002012665

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Use inverse function + antithetic variates in second choice

$$X \sim \text{Bernoulli}\left(\frac{1}{3}\right)$$

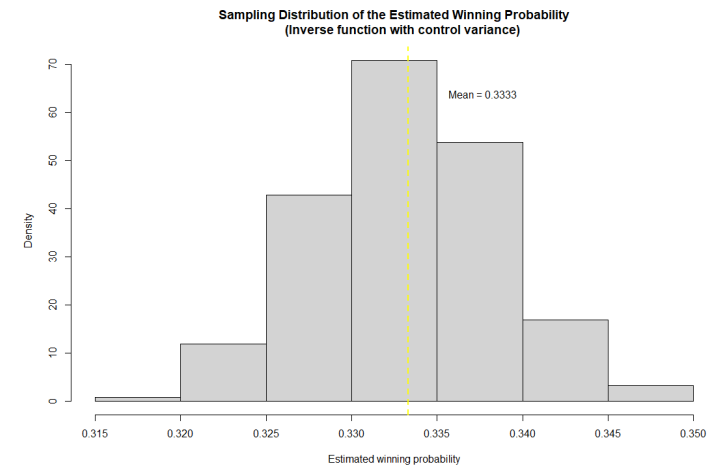
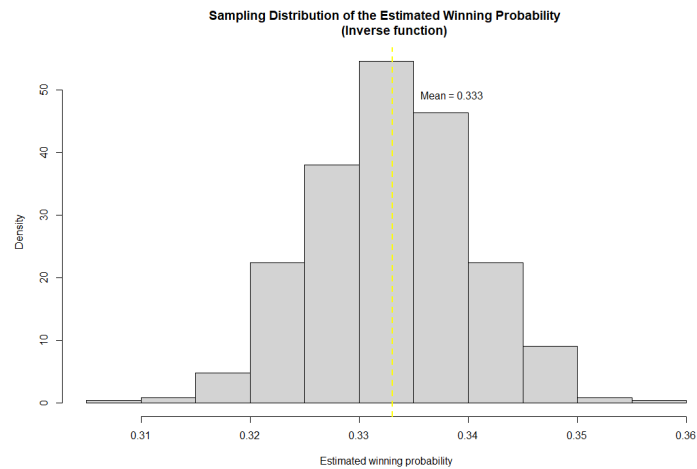
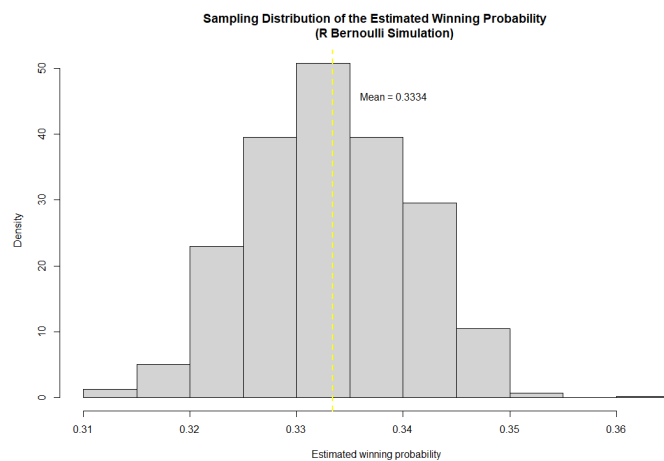
Inverse function

Antithetic variates

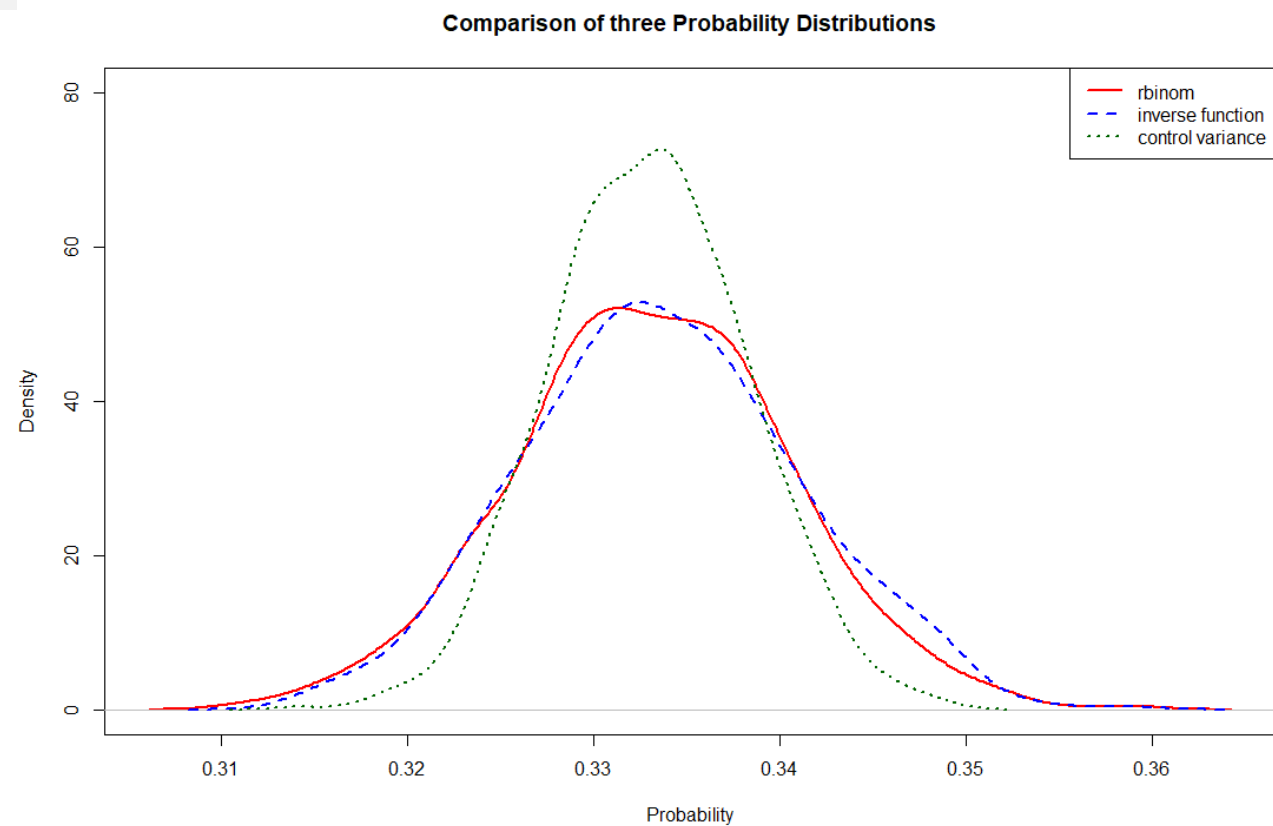
Antithetic variates

Antithetic variates

Antithetic variates



Comparison

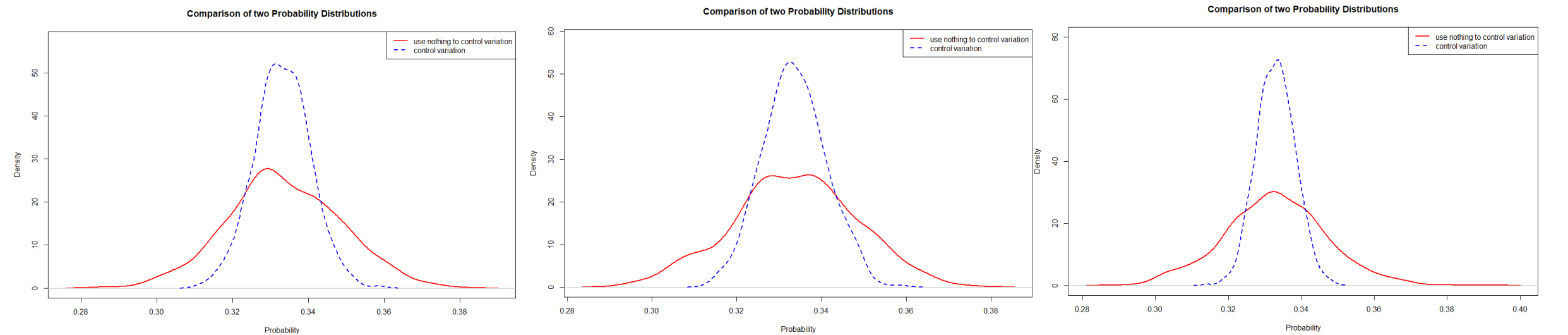


Variance reduction = 50.00826%

Method	Mean	Var
Built in R Bernoulli Simulation	0.3331201	0.00005636942
Inverse Function	0.3330853	0.00005416510
Inverse Function with Control Variance	0.3331452	0.00002888383

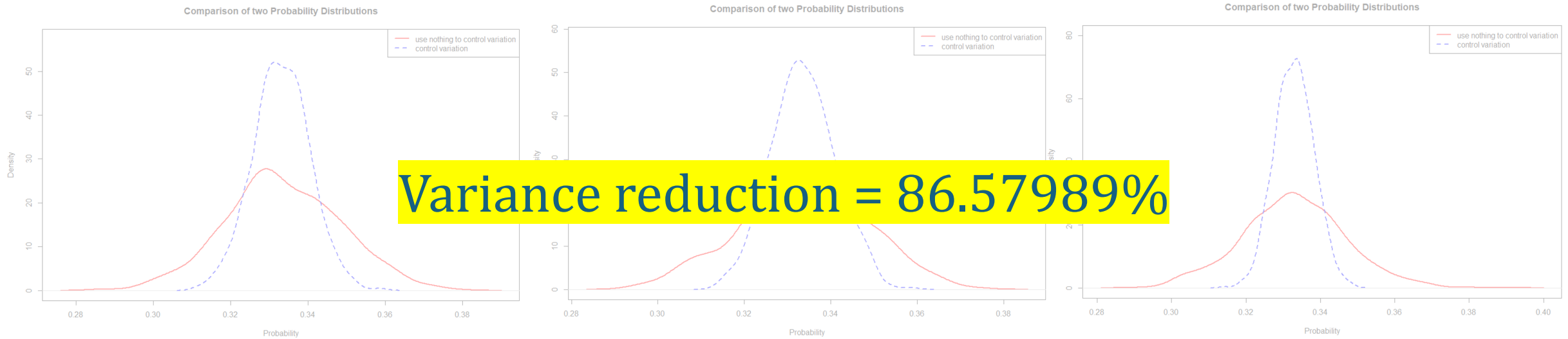
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Pairwise comparison



Method comparison			
—	Built in R Bernoulli + Built in R Bernoulli	Inverse + Inverse	Inverse Control+ Inverse
—	Built in R Bernoulli+ Inverse Control	Inverse + Inverse Control	Inverse Control+ Inverse Control
Variance reduction	75.70495 %	73.67488 %	85.64132 %

Pairwise comparison

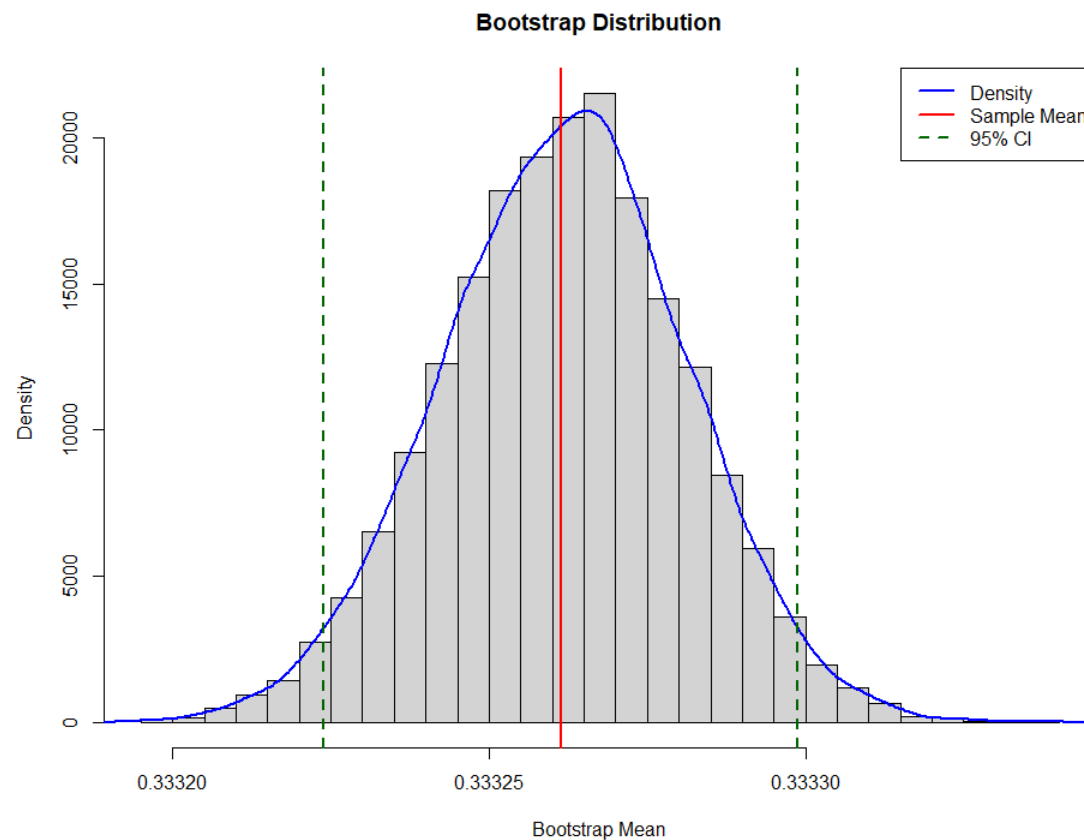


Method comparison			
—	Built in R Bernoulli + Built in R Bernoulli	Inverse + Inverse	Inverse Control+ Inverse
—	Built in R Bernoulli+ Inverse Control	Inverse + Inverse Control	Inverse Control+ Inverse Control
Variance reduction	75.70495 %	73.67488 %	85.64132 %

100個人
每個人玩了1000次
每次都換門的勝率



抽10000次

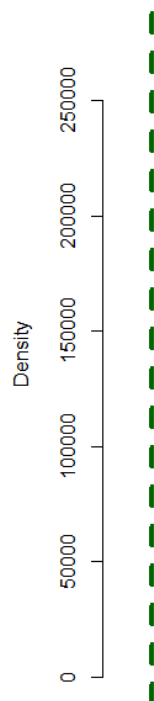


Method	mean	bias	std. error
Bootstrap	0.3332615	0.0000003272375	0.00001916393

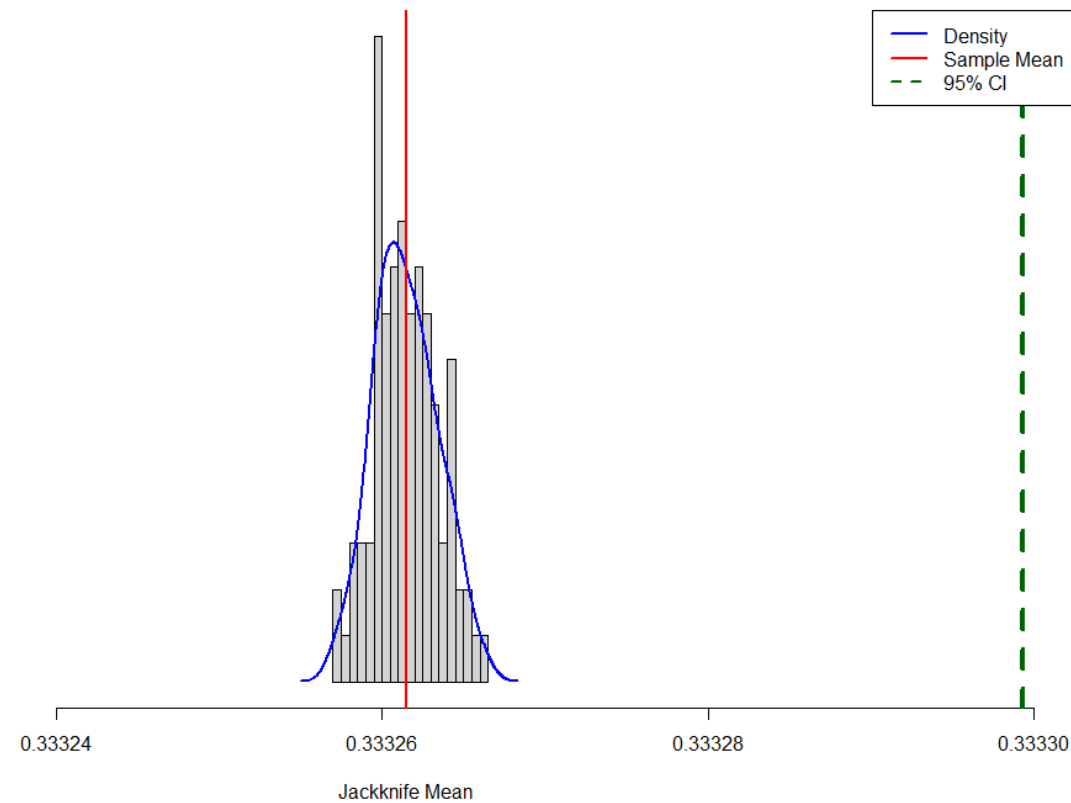
100個人
每個人玩了1000次
每次都換門的勝率



每次drop掉1人



Jackknife Distribution



Method	mean	std. error
Jackknife	0.3332615	0.00001929957

2 Statistical Inference - Permutation Test

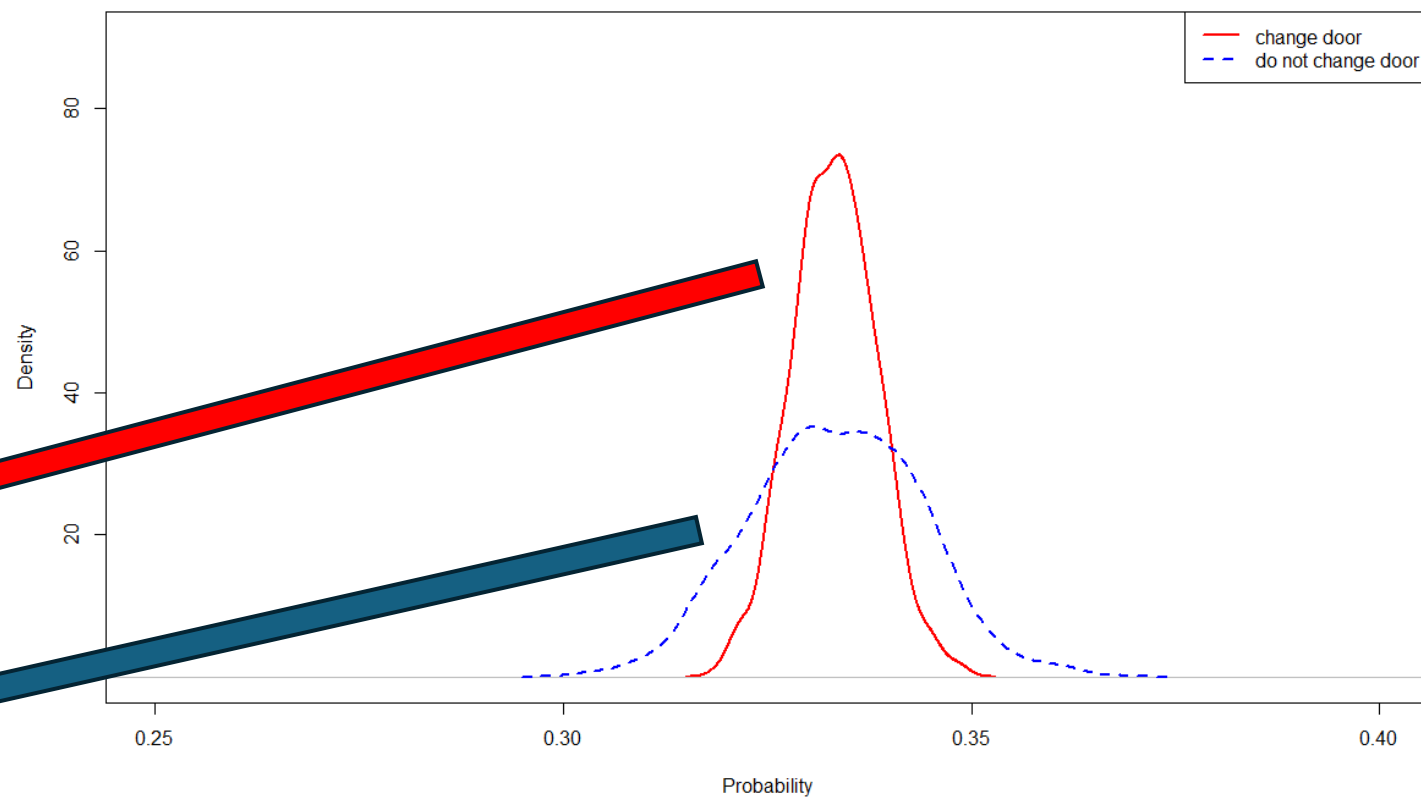
1000個人
每個人玩了1000次

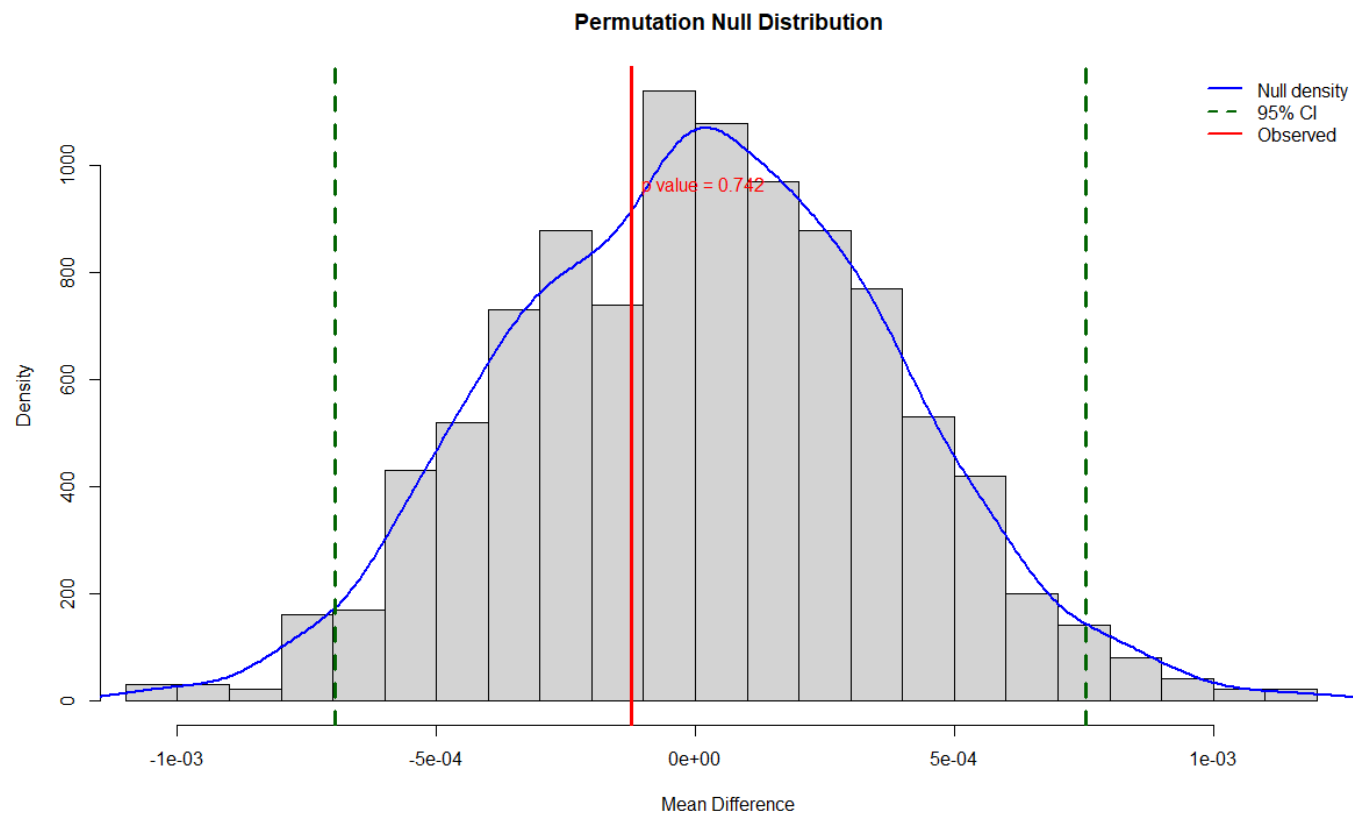
$$X_i = \begin{cases} 1, & \text{第一次選到跑車} \\ 0, & \text{第一次沒選到跑車} \end{cases}$$

$$Y_i = \begin{cases} 0, & X_i = 1 \\ Z_i, & X_i = 0, \quad Z_i \sim \text{Bernoulli}(\frac{1}{2}) \end{cases}$$

$$Y_i = \begin{cases} 1, & X_i = 1 \\ 0, & X_i = 0 \end{cases}$$

Comparison of two original Distributions





做了1000次

Method	mean	std. error	observed mean
Permutation	-0.00002076171	0.0003825296	0.000476954

1個人玩了1000次
每次都換門的結果

$$k = \sum_{i=1}^n y_i$$

$$P(\theta | y) \propto P(y | \theta)P(\theta)$$

$$\theta^k (1 - \theta)^{n-k}$$

$$\theta^{a-1} (1 - \theta)^{b-1}$$

根據原始情況，相信換門勝率高

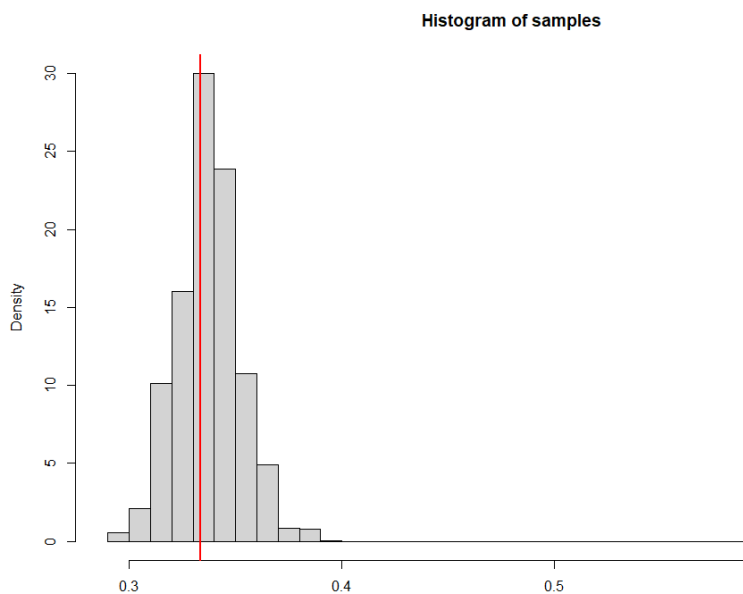
$$\theta \sim \text{Beta}(10, 5)$$

$$E[\theta] = \frac{10}{10 + 5} = \frac{2}{3}$$

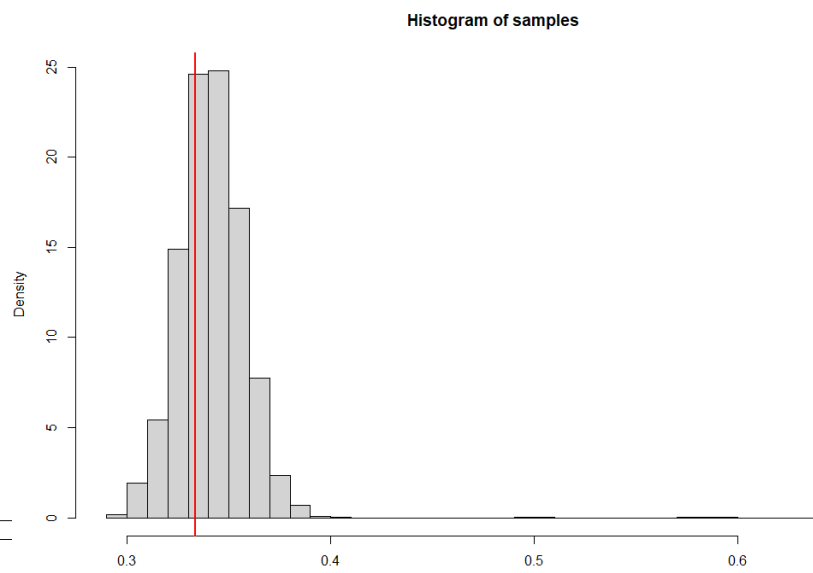
$$\theta^{k+a-1} (1 - \theta)^{n-k+b-1}$$

一開始認為換門勝率比較高，設定2/3

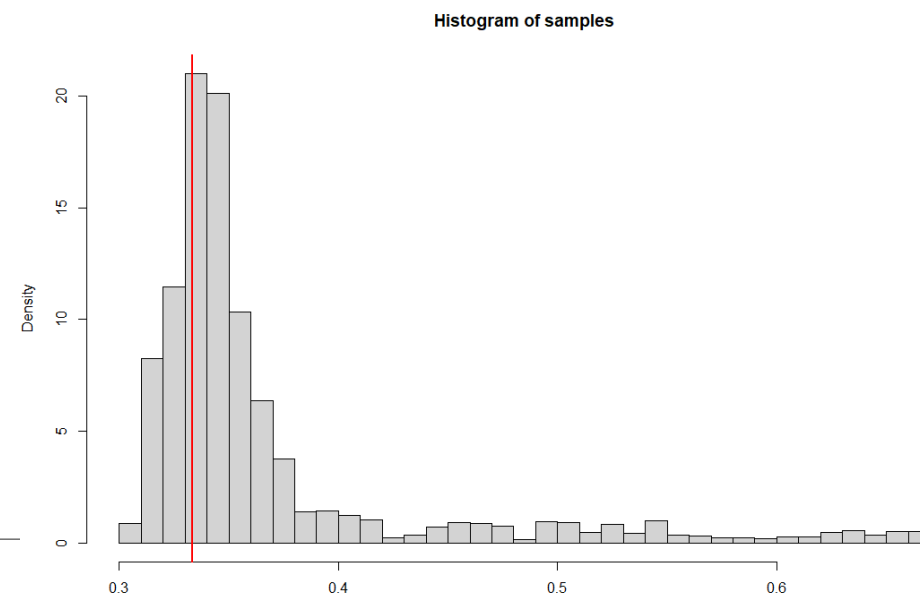
$$\theta_{\text{prop}} = \theta_{\text{current}} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$



$\sigma = 0.5$



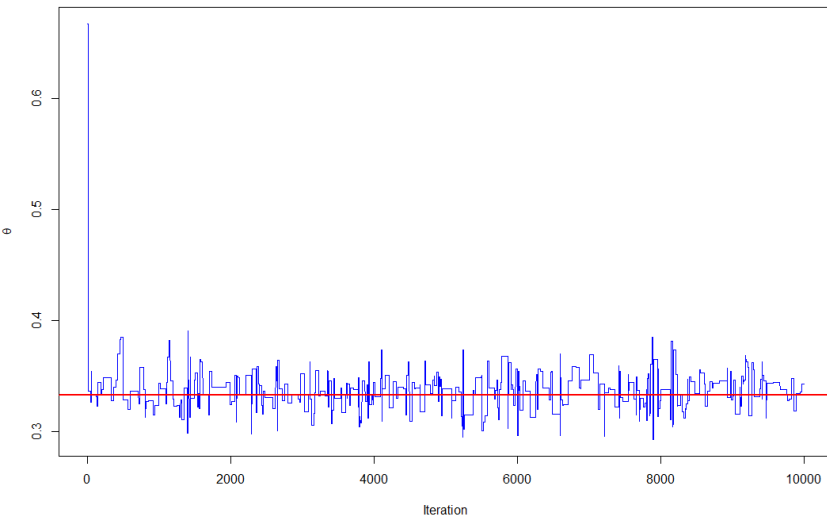
$\sigma = 0.05$



$\sigma = 0.001$

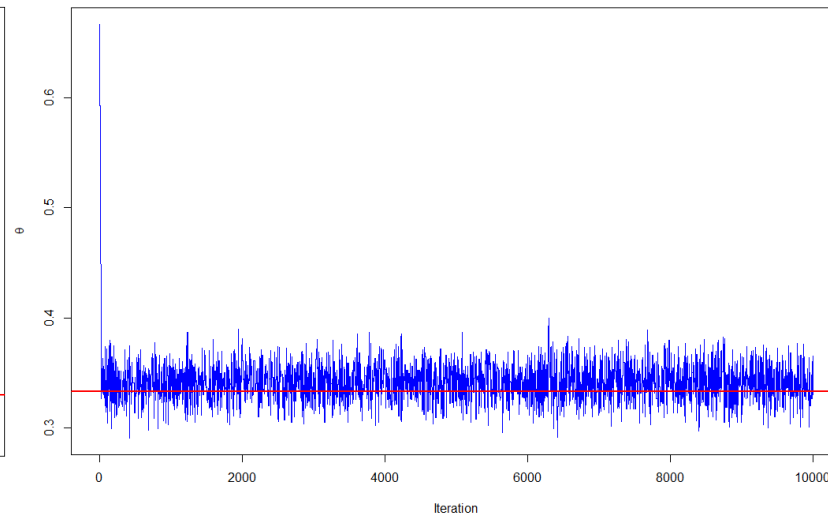
2 Statistical Inference: Markov Chain Monte Carlo (MCMC)-Metropolis Hastings Algorithm

MH Trace Plot



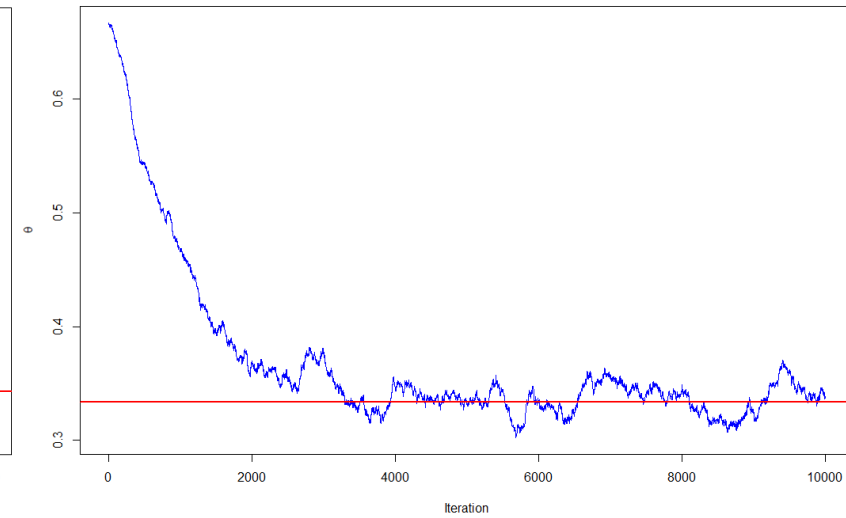
$\sigma = 0.5$

MH Trace Plot



$\sigma = 0.05$

MH Trace Plot



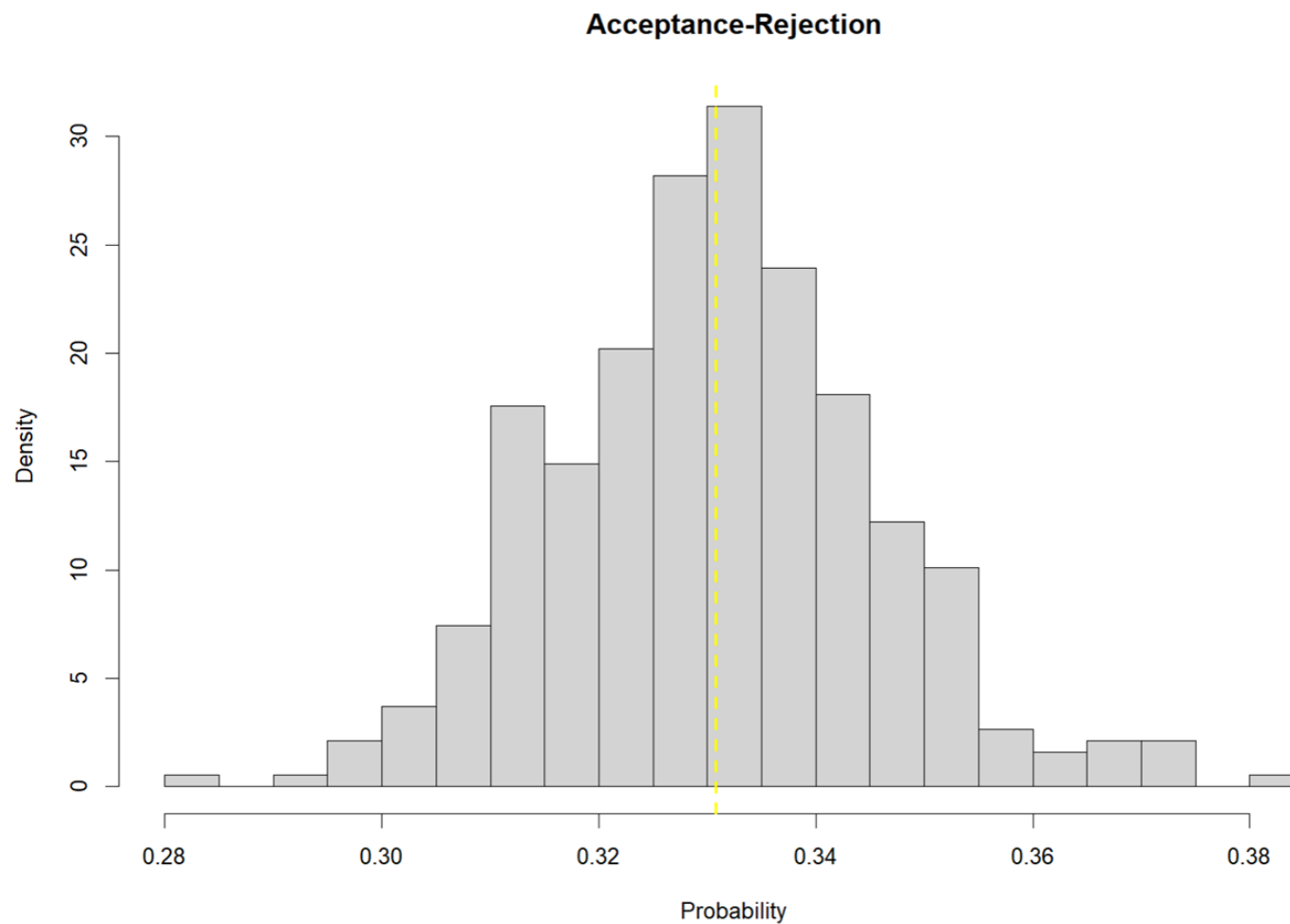
$\sigma = 0.001$

2 Statistical Inference: Acceptance - Rejection

$$g(t) \sim \text{Uniform}(0, 1)$$

$$\frac{f(t)}{g(t)} \leq c$$

$$N=10000$$

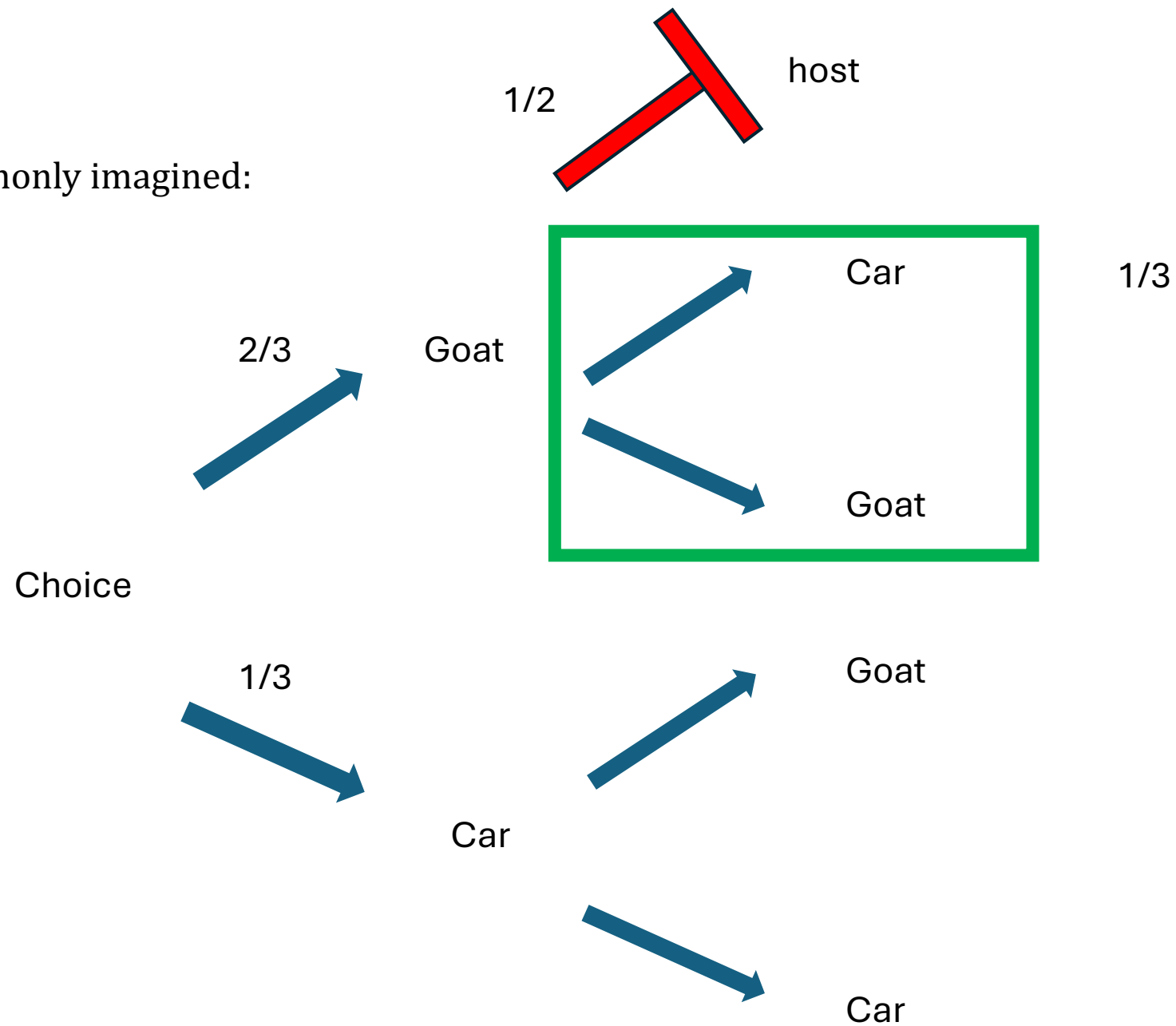




Explanation

- The situation that people commonly imagined:

$$P_{car} = 1/3$$





Explanation

- In fact, people ignored the situation which the host have the information:

$$P_{car} = 2/3$$

